THE LOCAL DIMENSION FORMULA DIM_U(U) - DIM_E(U)(R_U) IS INDEPENDENT OF PRESENTATION AND POINT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3189-3203 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 3205-3249, the complete author proof (43 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 7 of queue_stacks-properties.json (pb=1723, stmt 3189-3203, proof 3205-3249).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04TI: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 3249–3249. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-points-cartesian: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1019–1036. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NS: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 3249–3249. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AFI: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 3249–3249. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{X}$ be a locally Noetherian algebraic stack over a scheme S. Let $x \in |\mathcal{X}|$ be a point of $\mathcal{X}$. Let $[U/R] \to \mathcal{X}$ be a presentation (Algebraic Stacks, Definition 04TI) where U is a scheme. Let $u \in U$ be a point that maps to x. Let $e : U \to R$ be the "identity" map and let $s : R \to U$ be the "source" map, which is a smooth morphism of algebraic spaces. Let R_u be the fiber of $s : R \to U$ over u. The element*

$$\dim_x(\mathcal{X}) = \dim_u(U) - \dim_{e(u)}(R_u) \in \mathbf{Z} \cup \infty$$

is independent of the choice of presentation and the point u over x.

Proof. Since $R \to U$ is smooth, the scheme R_u is smooth over $\kappa(u)$ and hence has finite dimension. On the other hand, the scheme U is locally Noetherian, but this does not guarantee that $\dim_u(U)$ is finite. Thus the difference is an element of $\mathbf{Z} \cup \{\infty\}$.

Let $[U'/R'] \to \mathcal{X}$ and $u' \in U'$ be a second presentation where U' is a scheme and u' maps to x. Consider the algebraic space $P = U \times_{\mathcal{X}} U'$. By Lemma lemma-points-cartesian there exists a $p \in |P|$ mapping to u and u'. Since $P \to U$ and $P \to U'$ are smooth we see that $\dim_p(P) = \dim_u(U) + \dim_p(P_u)$ and $\dim_p(P) = \dim_{u'}(U') + \dim_p(P_{u'})$, see Morphisms of Spaces, Lemma 0AFI. Note that

$$R'_{u'} = \mathrm{Spec}(\kappa(u')) \times_{\mathcal{X}} U' \quad \text{and} \quad P_u = \mathrm{Spec}(\kappa(u)) \times_{\mathcal{X}} U'$$

Let us represent $p \in |P|$ by a morphism $\mathrm{Spec}(\Omega) \to P$. Since p maps to both u and u' it induces a 2-morphism between the compositions $\mathrm{Spec}(\Omega) \to \mathrm{Spec}(\kappa(u')) \to \mathcal{X}$ and $\mathrm{Spec}(\Omega) \to \mathrm{Spec}(\kappa(u)) \to \mathcal{X}$ which in turn defines an isomorphism

$$\mathrm{Spec}(\Omega) \times_{\mathrm{Spec}(\kappa(u'))} R'_{u'} \cong \mathrm{Spec}(\Omega) \times_{\mathrm{Spec}(\kappa(u))} P_u$$

as algebraic spaces over $\mathrm{Spec}(\Omega)$ mapping the Ω-rational point $(1, e'(u'))$ to $(1, p)$ (some details omitted). We conclude that

$$\dim_{e'(u')}(R'_{u'}) = \dim_p(P_u)$$

by Morphisms of Spaces, Lemma 04NS. By symmetry we have $\dim_{e(u)}(R_u) = \dim_p(P_{u'})$. Putting everything together we obtain the independence of choices. □